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Air for Your Engine

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Air is as necessary to an internal combustion engines as is fuel. The way air is used in an engine can make a lot of difference in operating costs. All internal combustion engines use air, and all of them can be damaged in the same way by its misuse.

Your operating costs will be lower; you will save on fuel and repair costs, and your engine will perform better if you make provisions for your engine to have a Sufficient Volume of Clear Air at the Right Temperature.

Your Engine **must** have lots of air

The engine **must** have a sufficient volume of air to convert all its fuel into power instead of smoke. Smoke coming out of the exhaust stack is wasted fuel, and the waste can be as much as 15 % of the total fuel cost.

Black exhaust smoke coming from an engine means that it is **not** receiving enough air to burn all of its fuel. **Not** all of this unburned fuel goes out the exhaust stack; some of it washes down the cylinder walls to dilute the lubricating oil, or it changes to carbon which can stick to the piston rings or clog the injector spray holes.

Clean Air

- Dirt is the engine's worst enemy. It has been proven many times that the way to get extra service out of an engine is to keep the intake air clean.
- All air contains small particles of dirt and abrasive material. Dirty air is always present but **not** always visible. On some job sites the air can be so dirty that you will ruin an engine in a short period of time. You **must** take proper measures to clean the air.
- There has never been an engine designed that will give satisfactory service if it is allowed dirty unfiltered air. This is why the job of the air cleaner is so important and why it is necessary to keep it checked and serviced frequently.
- Dirty intake air is the main cause of wear on pistons, rings, liners, valves and many other parts.
- For instances of dust/dirt ingestion, see Service Bulletin, High Blowby and Lubricating Oil Consumption Caused by Dirt and Dust Ingestion, Bulletin 5613318 (</qs3/pubsys2/xml/en/bulletin/5613318.html>).

Air Temperature

- Diesel engines are best suited for air temperatures between 60 and 90°F [15 and 32°C]. Engines can withstand temperatures below or above this range, but their efficiency drops.

Intake Air that is too hot

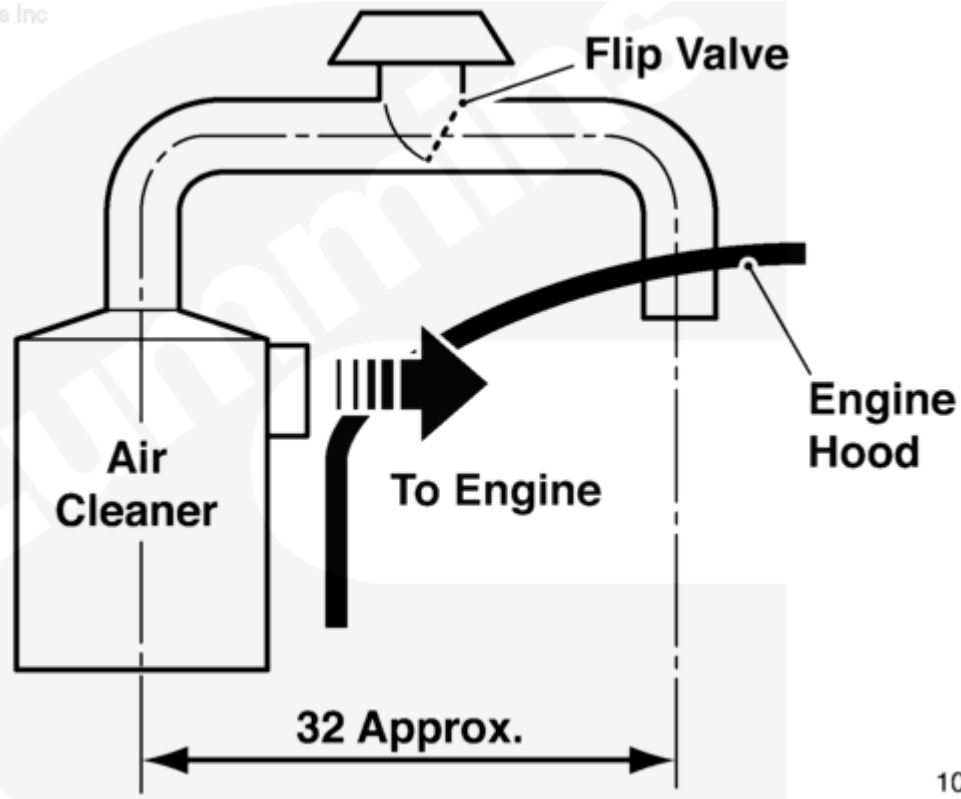
- Engine horsepower falls about 1% for each 10 degrees of intake air temperature rise above 90°F [32°C].
- An engine rated at 250 horsepower will develop only 240 horsepower when the intake air temperature is 130°F [54°C] with the same fuel delivery.
- Cummins Inc. engines are designed to work with OEM supplied charge air coolers. charge air coolers cool down the compressed(charged) air after turbocharger, before air reaches the air intake manifold.

Air that is too cold

- Cummins Diesels are rated on the basis of intake air at 85°F [29°C] temperature, but in most localities engines operate part of the time at temperatures of freezing or below. A drop of 60 degrees in intake air temperature results in a 160-degree drop in compression temperature.
- The knocking of a diesel engine when you first start it in cold weather is a warning that the compression temperature is too low.
- Low compression temperatures have the following effects on engine operation:
 1. Failure to ignite the fuel
 2. Delay in ignition of the fuel and, in turn, detonation or fuel knock, and higher peak combustion pressures
 3. Irregular combustion and rough-running engine, particularly at idling speeds
 4. Possible periodic failure to ignite the fuel charge with resultant dilution, lubrication problems, ring troubles, etc.

Controlling the Air Temperature

- Some equipment manufacturers and operators have recognized the advantages of controlling the temperature of the intake air and have devised methods to provide an alternative source when the intake air temperature is abnormally high or low.
- Consideration **must** be given to high temperatures resulting from enclosed engines such as those in compartments on boats or enclosure for stationary power units. In these situations an arrangement is needed to bring cooler outside air to the engine.
- Similarly, some equipment manufacturers have designed methods of drawing warmer air from under the hood for engines operating in cold climates. These arrangements normally will have provisions for changing to outside air during hot weather



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Exterior Mounted Air Cleaner with Flip Valve

Note : The ideal intake air temperature for most efficient operation of a diesel engine is between 60 and 90°F [15 to 32°C].

Providing Air for your Engine

- You **must** do what is necessary to provide the amount and kind of air the engine needs.
- Restrictions in the air inlet will prevent your engine from receiving the air it needs for efficient operation.

Measure Air Inlet Restriction

- Measure the air inlet restriction with a manometer at one to two pipe diameters upstream from the turbocharger inlet in a straight section of pipe or at the intake manifold on naturally aspirated engines.
- If the restriction is lower than the maximum allowed at governed speed, (turbocharged engines **must** be under full load), the engine is getting enough air.
- If the restriction on a turbocharged engine exceeds 25 inches [635 mm] of water, or 1.8 inch [46 mm] of mercury, the engine is **not** getting enough air.
- Restriction on a naturally aspirated engine **must not** exceed 20 inches [0.5 m] of water or 1.5 inches [38 mm] of mercury. To correct, proceed as follows:
 1. Clean the air cleaner screens as described under "Providing Clean Air For Your Engine"
 2. Check for sharp bends in the air inlet piping. Sharp bends restrict air flow
 3. Check to be sure the inside diameter of the piping between the cleaner and the engine, or of the extension or "snorkel" above the air cleaner, is no smaller than the flange connections of the air cleaner
 4. Repair or replace any dented or mashed air cleaner piping, rain shield or air cleaner

5. When replacements are made, make sure the air cleaners are of adequate size and capacity.

- After all corrective measures have been taken, recheck the inlet air restriction.

Reduced Load For High Altitude Operation

- At high altitudes, the air is too thin for the engine to burn as much fuel as it can at sea level, and you should operate in a lower gear or cut back the fuel to avoid a smoky exhaust.

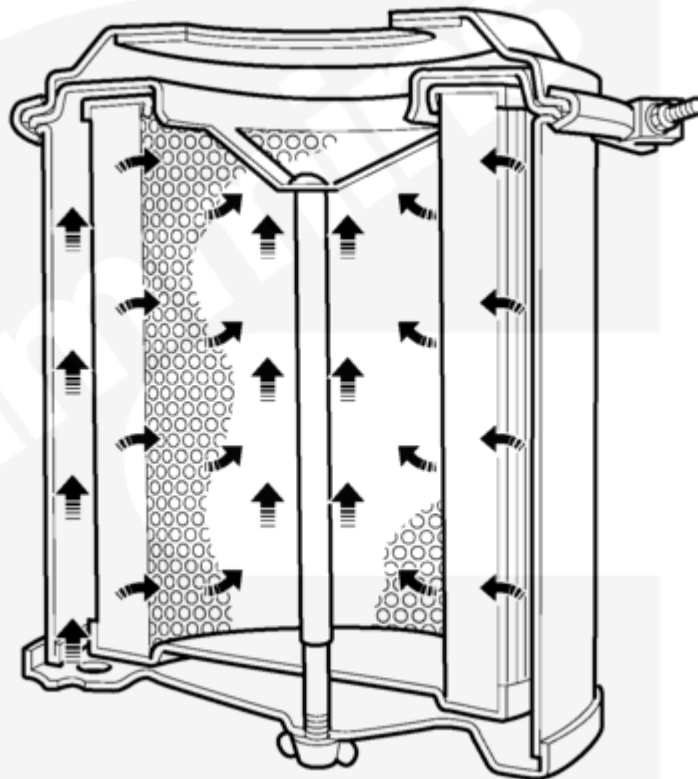
Providing Clean Air For Your Engine

- An air cleaner is used to remove dirt and dust from the air your engine intakes. Since dirt is an engine's worst enemy, it is important to check and service the air cleaner regularly.

Types Of Air Cleaners

- The best air cleaner is the one that supplies an engine with all the air it needs — all of the time — and, at the same time, keeps harmful abrasives from entering with the air.
- Laboratory tests have proven that an air cleaner with a 98% efficiency rating will pass 200 times as much dirt as a cleaner with 99.99% efficiency rating.
- Cummins Inc. recommends the use of the following basic types of air cleaners.

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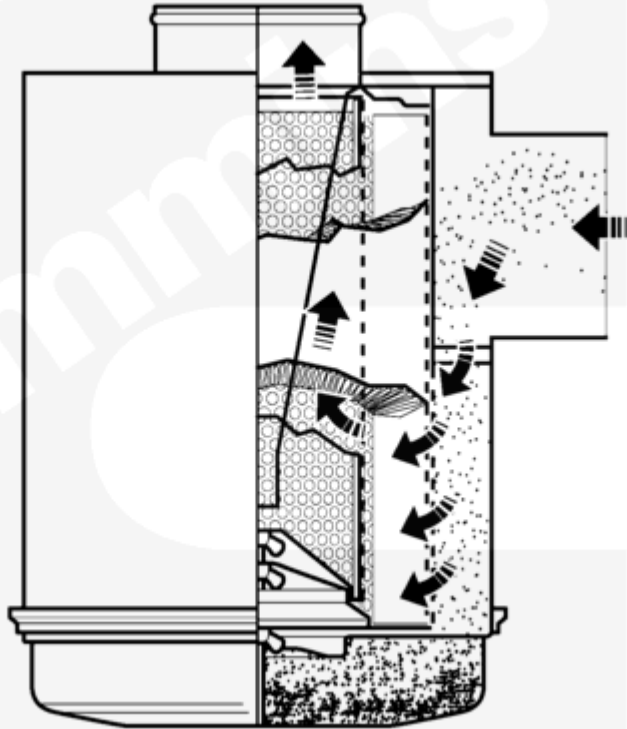
Dry Type Air Cleaner

A dry-type air cleaner is a positive filtration type consisting of treated paper as a medium that air passes through to reach the engine.

The dry-type air cleaner increases in efficiency as the dirt load builds up a cake or bed in the filtered medium, upwards from a minimum efficiency of 99.5%. This efficiency is constant at all engine speeds.

Intake systems **must** have some means of removing water before it gets to the dry-type cleaner element, especially on-highway trucks which are subjected to water-salt spray and splash solutions from the highway. Inline and stack type separators are available for removing water and water-salt spray from intake air.

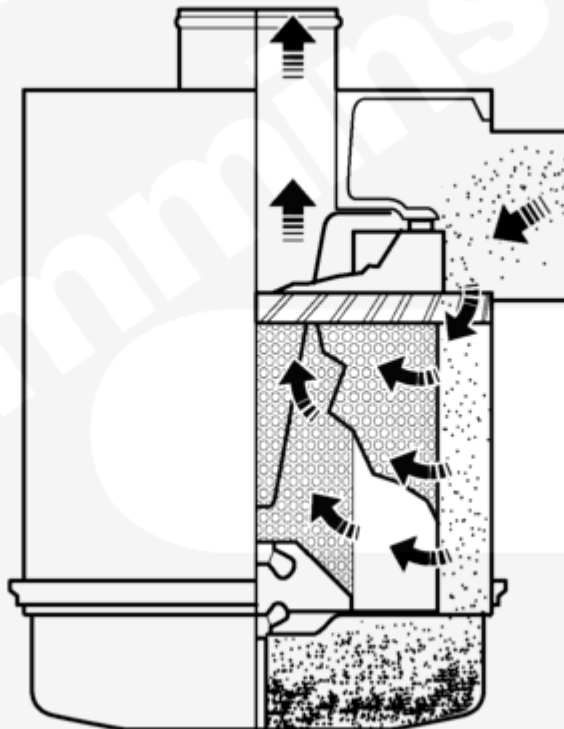
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Composite Dry Type Air Cleaner - Dual Element

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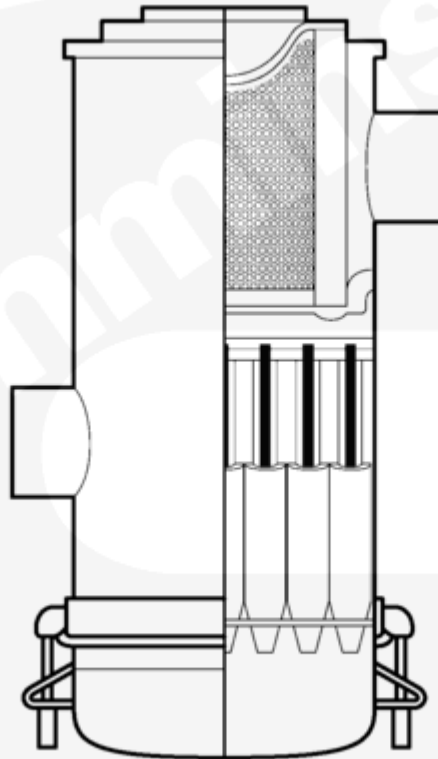
Composite Dry Type Air Cleaner - Single Element

Composite cleaners utilize centrifugal force and a paper filter for efficient two-stage cleaning. These are available in cylindrical or panel type air cleaners.

On the composite cylindrical model, dirty air enters through the inlet opening, where it immediately travels through a ring of vanes around the outside of the element. These vanes create a cyclonic twist to the air, thereby throwing dust and dirt particles outward and down into the dust cup.

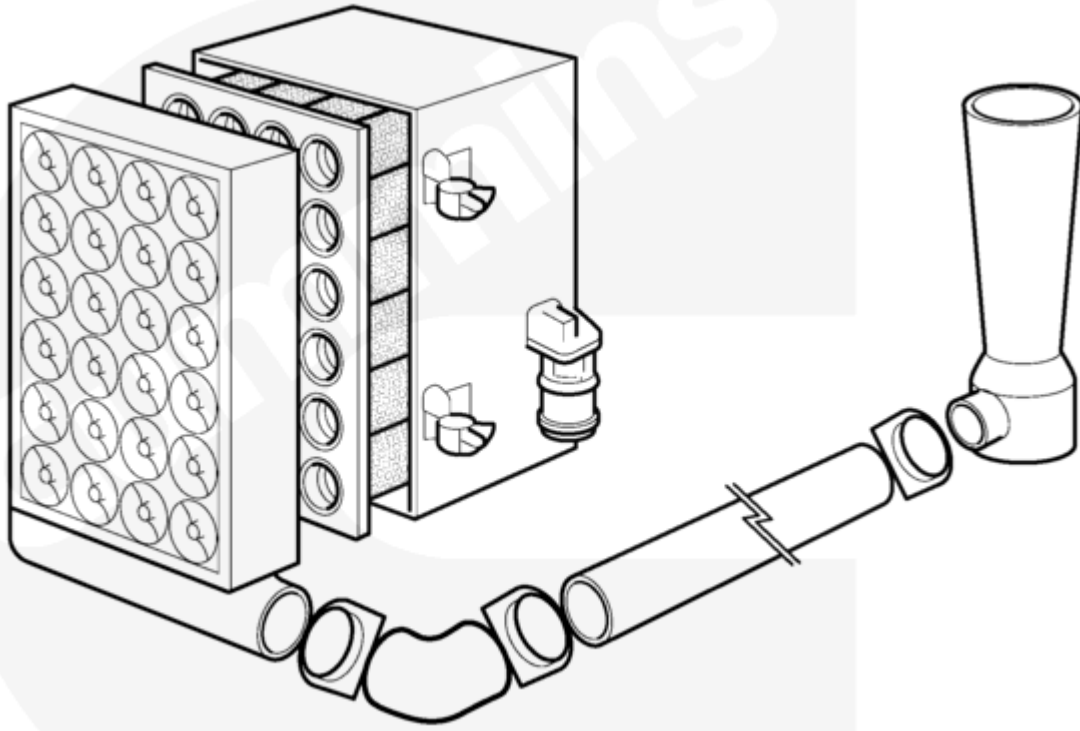
On the composite tube type, dirty air enters through the inlet and passes into a series of vaned tubes which creates a cyclonic twist to the air. Dust particles, separated by this twist action and thrown to the outside of the tube, falls into a removable dust cup. The clean air passes through the return tubes into the paper filter which removes the remaining dust.

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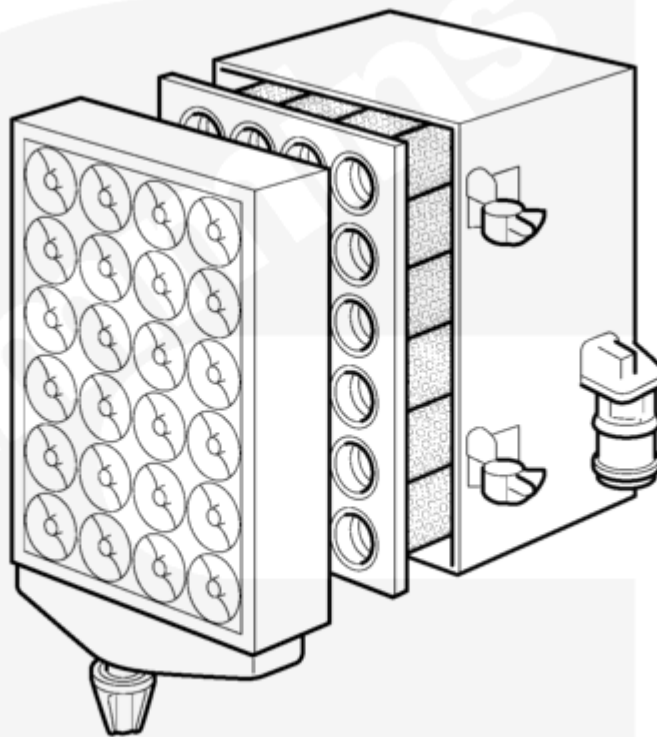
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Composite Dry Type Air Cleaner - Tube Type



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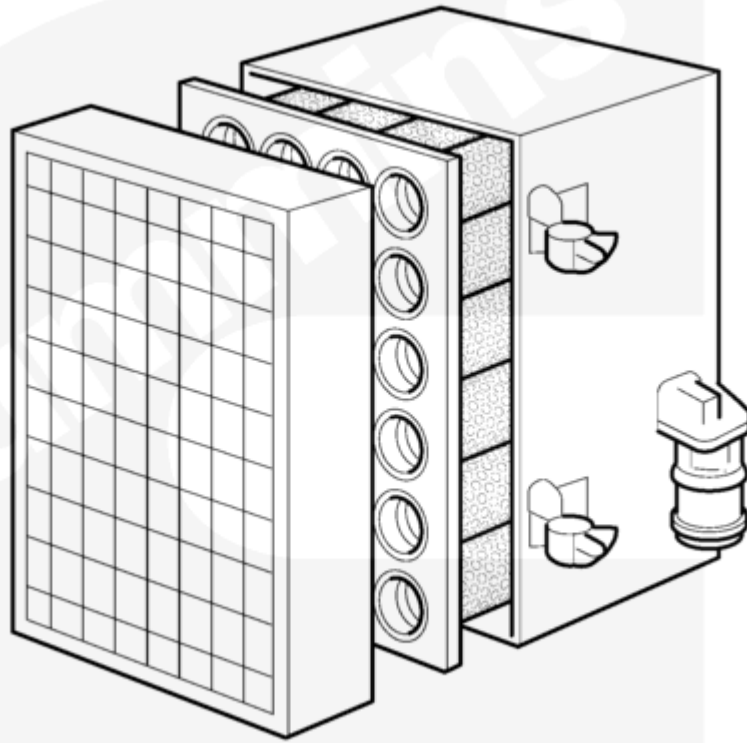
Cartridge Type Air Cleaner - Two Stage



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Cartridge Type Air Cleaner - Two Stage Without Exhaust Aspirator

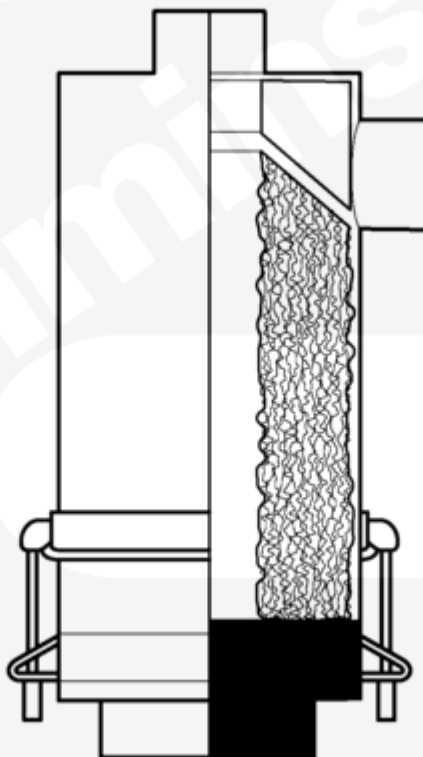
The two-stage cartridge type air cleaner has deflector vanes which create a cyclonic twist as the air enters. This removes most of the dirt which drops into the dust bin. The air reverses direction and then spirals back into the air cleaner element. An exhaust aspirator can be used with this type of air cleaner.



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Cartridge Type Air Cleaner - Single Stage

The single stage cartridge type air cleaner is equipped with a moisture eliminator. The moisture eliminator is fitted in front of the cartridge to attract moisture and dust. The coils of the moisture eliminator attracts moisture where it drains and is expelled at the base of the eliminator. The air is further cleaned by being pulled through the paper filters.



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Oil Bath Air Cleaner

An oil bath air cleaner contains a pool of oil and a wire mesh medium. As the air is drawn over the oil pool, a portion of the abrasives attach themselves to the oil. Further filtering occurs as the air stream pulls oil onto the mesh. Efficiency is up to 98.5% at governed speed with a lower efficiency at idle speeds.

Recommendations

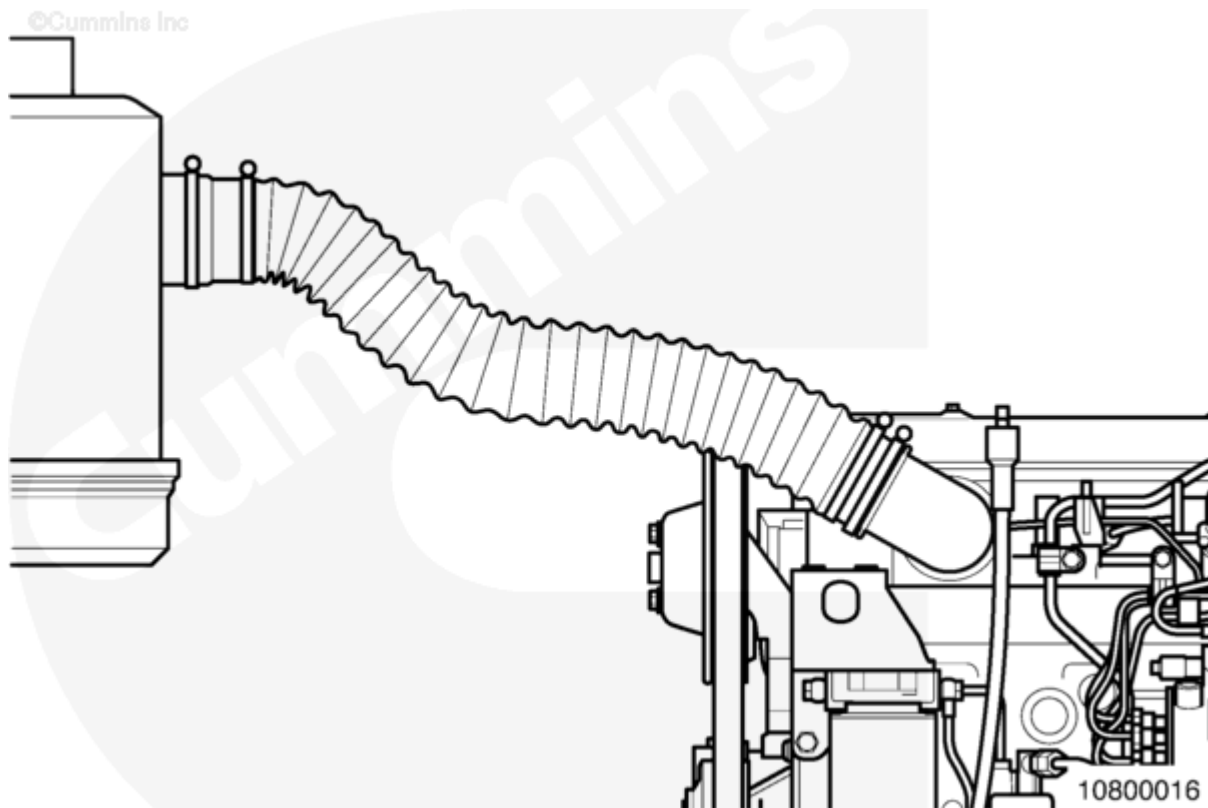
- The dry type air cleaner is recommended for use under light dust conditions. Composite and cartridge type air cleaners are recommended for use in medium and heavy dust conditions.

Maintenance

- Air cleaners **must** be maintained to operate efficiently. For complete instructions on performing maintenance checks consult the Engine Operation and Maintenance Manual for your particular engine model and application.
- If you have problems regarding selection, installation, or maintenance of air cleaners for your operation, please contact your Cummins Distributor for advice.

Tubing, Hose And Clamps

- Air induction piping works together with the air cleaner in carrying air to your engine. it is important that all the joints are properly sealed and free of leaks. An air cleaner is completely ineffective if leaks occur in the piping between the air cleaner and the engine.



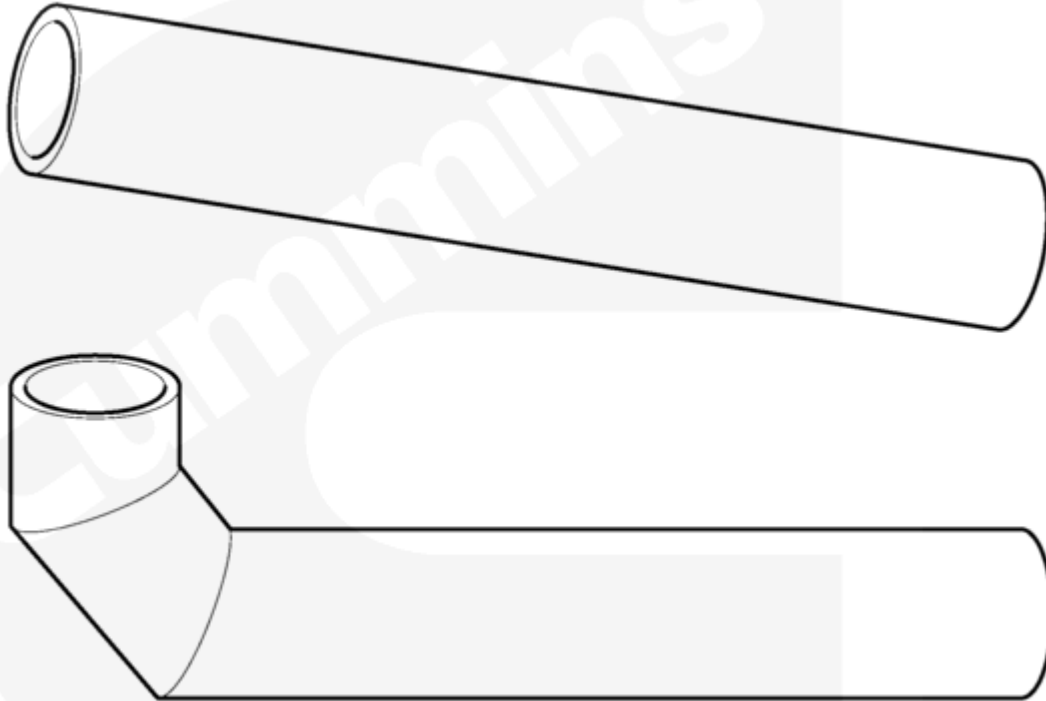
Wire-Reinforced Hose

Dirt is the basic cause of wear of pistons, rings, liners and valves. One of the most probable places for dirt to enter an engine is through an opening in the air induction piping.

Field experience has shown that most air leaks occur when wire-reinforced hoses are used in the air induction system. These leaks are caused by wire wearing through the hose fabric and often are barely visible to the naked eye. A very small hole can allow large quantities of dirty air to enter an engine.

Cummins Inc., recommends the use of the following tubing, elbows, connections and clamps as the best assurance of dirt-free air induction.

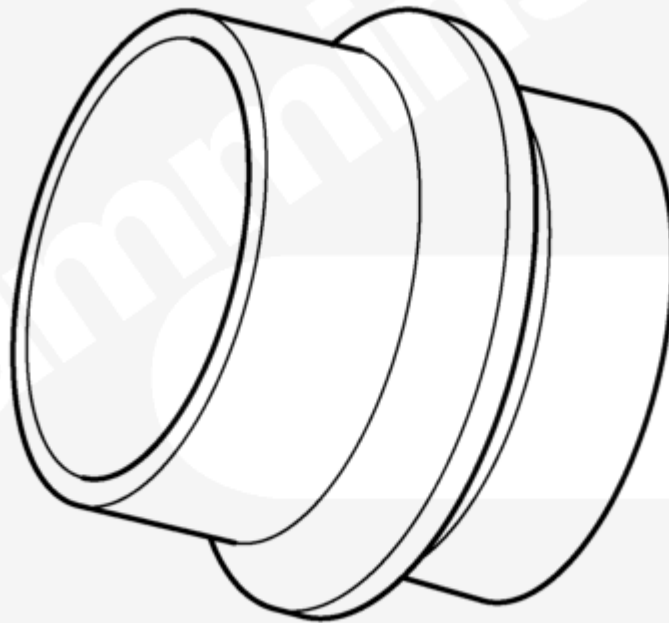
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Smooth-Welded Steel Tubing

Use smooth-weld steel tubing instead of flexible hose or metal tubing with rough weld steel tubing angle joints. This steel tubing provides a smooth surface for sealing with a rubber hose.

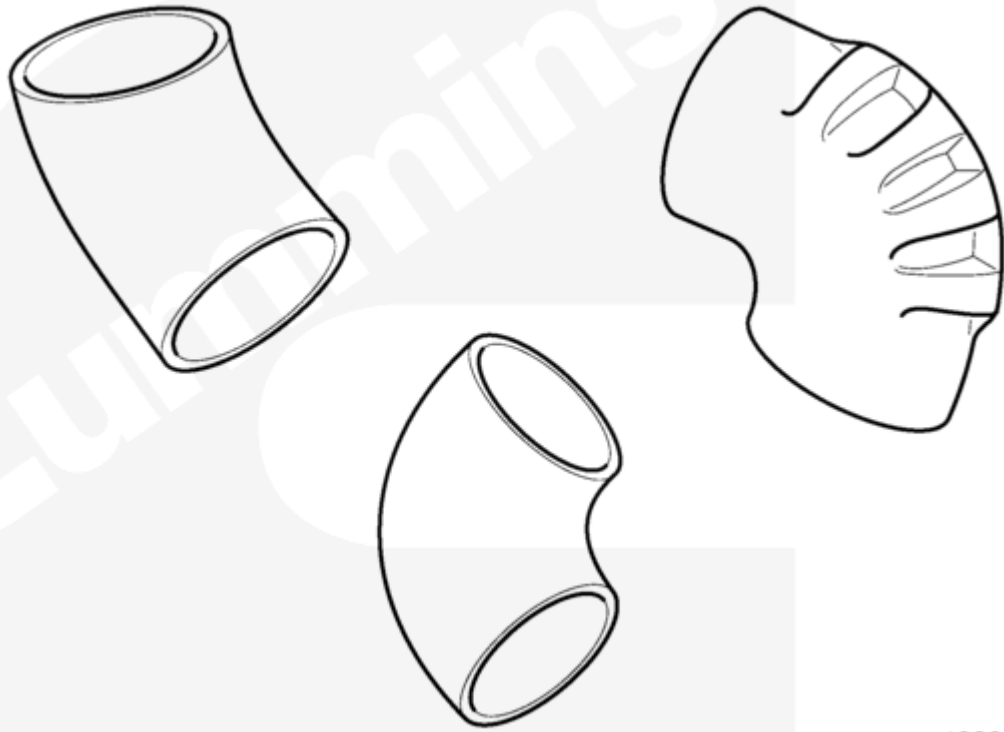


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Connecting Hose

Specially developed “hump” hoses provide airtight connections and are flexible enough to absorb shocks transmitted by steel tubing. The “hump” permits angular and axial movement and prevents collapse of the hose when the connection tubing is too far apart. It also protects the hose from being cut when the tubes are too close together.

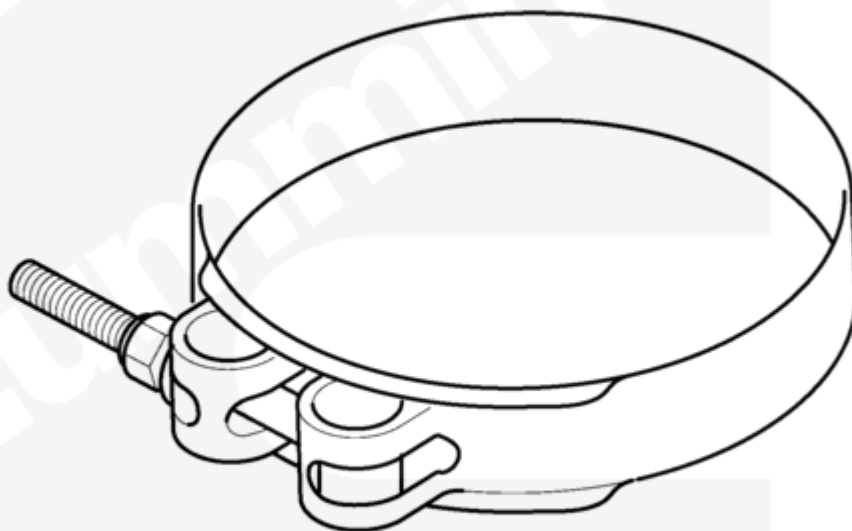
Note : Tubing must be spaced at least 3.4 inch [19.05 mm] apart when connected by a hose.



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Molded Rubber Elbows

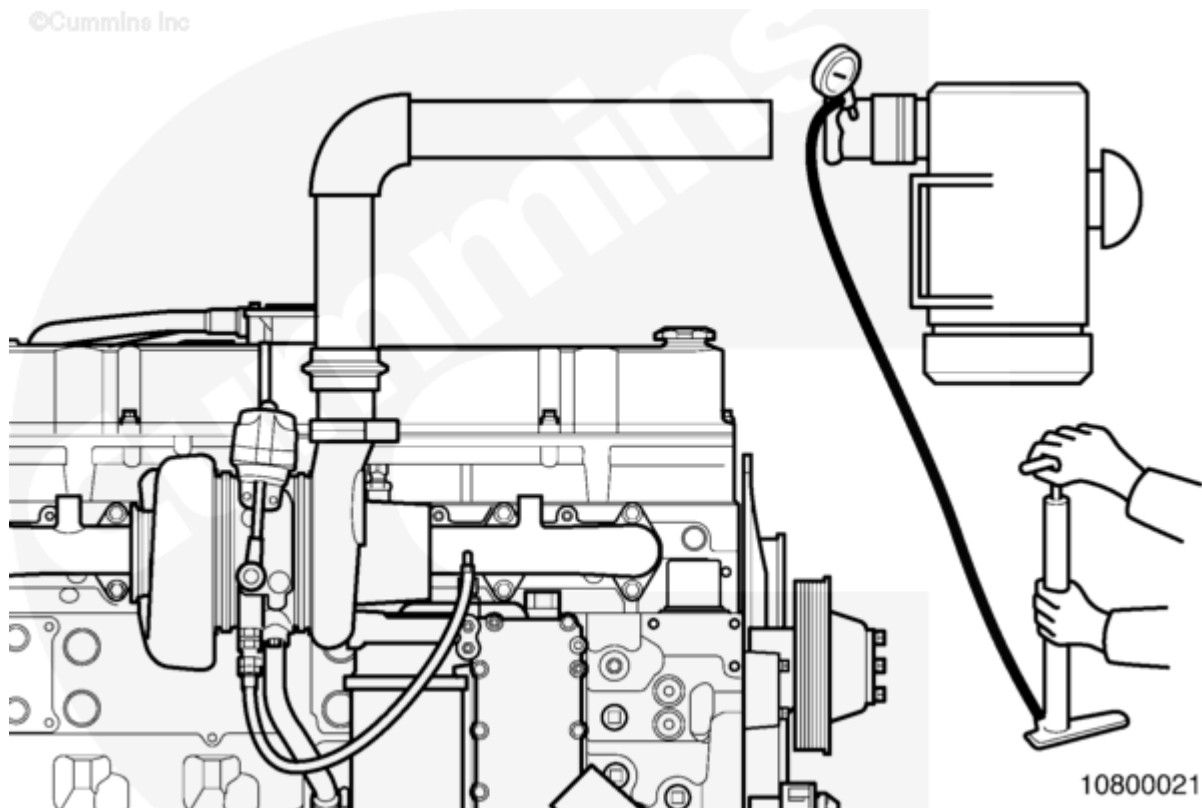
Molded rubber elbows have been designed to replace flexible rubber hose and welded angle joints. They are available in both 90 degree and 45 degree angles with a wall thickness of 1/4 inch [6.35 mm]. A 90 degree elbow with "ribbed" reinforcement is also available; it is designed to prevent collapsing under conditions of high temperature or high inlet restriction.



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Hose Clamps

For connection purposes, "T" bolt type, noncrimping hose clamps are recommended. When tightened, equal pressure is exerted around the circumference of the hose. A locknut prevents the clamp from slipping, thus assuring a perfect and permanent seal.



Pressure Checking Air Leaks — Turbocharger Engine

Pressure checking is a good way to test the engine air intake system for leaks. The purpose of pressure checking is to make sure that all the air going to the engine passes through the air cleaner.

All leaks will be found when pressure checking is used properly and a sufficient amount of liquid soap has been applied to the exterior of the air induction system.

Check starting aids to make sure contaminants can **not** enter through this source.

The general principles of pressure checking are the same for all Cummins engines. However, the actual details vary according to the type of air cleaner used and whether the engine is naturally aspirated, supercharged or turbocharged.

Clean the dirt off the clamps, hose and piping before removing them to keep dirt from entering the air intake system.

Note : The same pipe tap hole used for pressure checking can be also be used to install the manometer for checking air inlet restriction.

Turbocharged Engine With Oil Bath Cleaner (1)

- a. Clamp a rubber sheet tightly over the air cleaner inlet
- b. Seal the air cleaner body-to-oil cup parting line with rubber tape if the cleaner oil cup is **not** equipped with an o-ring seal.

- c. Disconnect the air cleaner outlet piping at the turbocharger connection and clamp a rubber plug over the piping to provide an air tight seal at the turbocharger end
- d. Install the pressure gauge in the piping.
- e. Apply liquid soap to the exterior of the air cleaning system and then apply air pressure. Check for bubbles and correct leaks.

Note : It may be necessary to drill and tap an 1/8 inch hole for the fittings. Where it is not desirable to drill a hole, clamp the valve stem section of an inner tube over the air cleaner inlet, and then apply air through the valve stem.

Turbocharged Engines With Dry-Type Cleaner (2)

- a. Remove the cleaner element and install one which is pre-wrapped with masking tape to prevent the passage of air.
- b. Tighten the tape-covered element in place
- c. Proceed with the air check as described in (1) above, steps c, d, and e.

Note : It is suggested that a pre-wrapped element be kept in stock for pressure checking. A new element is desirable, but an undamaged used element will perform satisfactorily.

Naturally-Aspirated Engines With Oil Bath Cleaner (3)

- a. Repeat (1) above, steps a and b
- b. Insert a sheet metal plate or dummy gasket between the intake manifold and the air inlet connection
- c. Install a pressure gauge in the intake manifold
- d. Apply liquid soap to the exterior of the air cleaner system, and then apply the air pressure. Check for bubbles and correct the leaks.

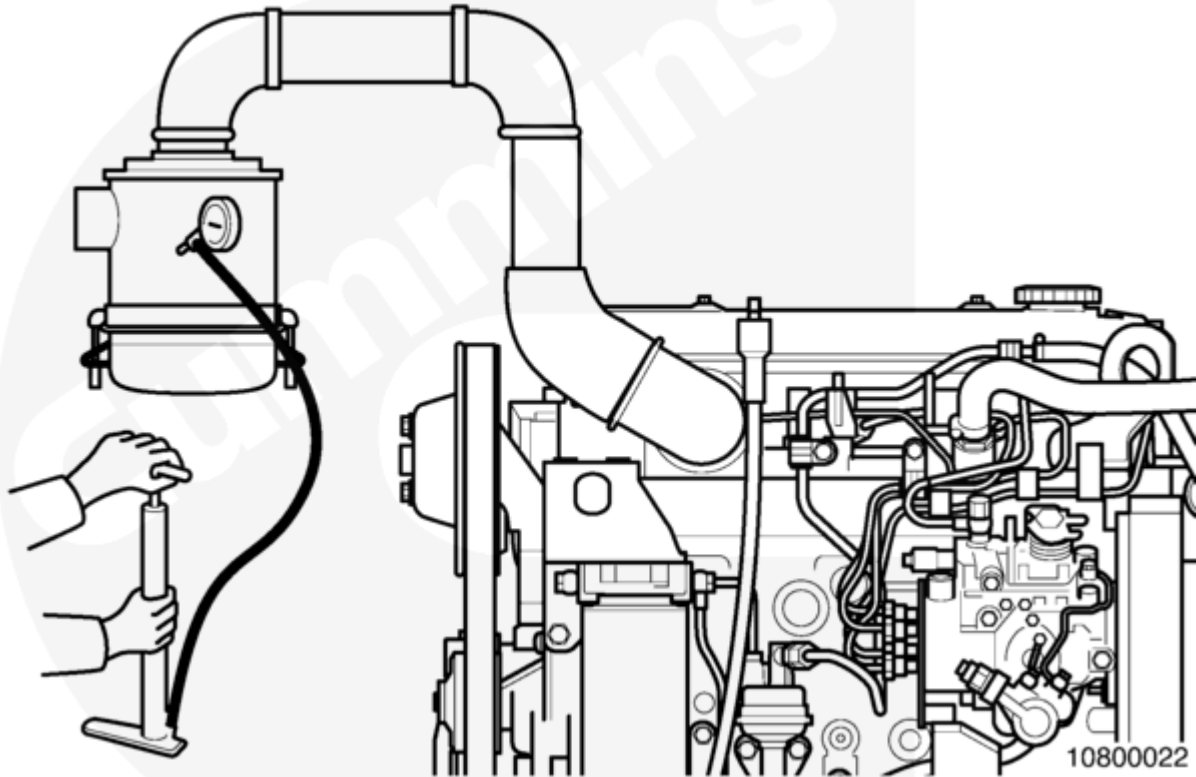
Naturally-Aspirated Engines With Dry-Type Cleaner (4)

- a. Remove the cleaner element and install one that is pre-wrapped with masking tape to prevent passage of air.
- b. Tighten the tape covered element in place
- c. Apply liquid soap to the exterior of the air cleaning system, and then apply the air pressure. Check for bubbles and correct all leaks.

Note : It is suggested that a pre-wrapped element be kept in stock for pressure checking. A new element is desirable, but an undamaged used element will perform satisfactorily.

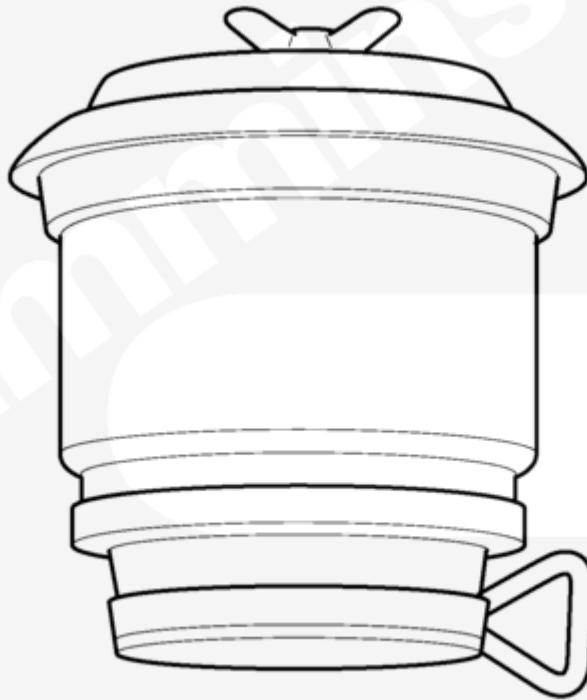
All Engine With Composite Tube Type Air Cleaner (5)

- a. Attach a pressure gauge and an air pressure valve at the tap connection in the cleaner body
- b. Remove the cleaner cover and the element. If an inner cover is used, remove it also
- c. Install an outlet plug or plate and gasket arrangement to the air cleaner outlet. The kits are **not** available from Cummins Inc.



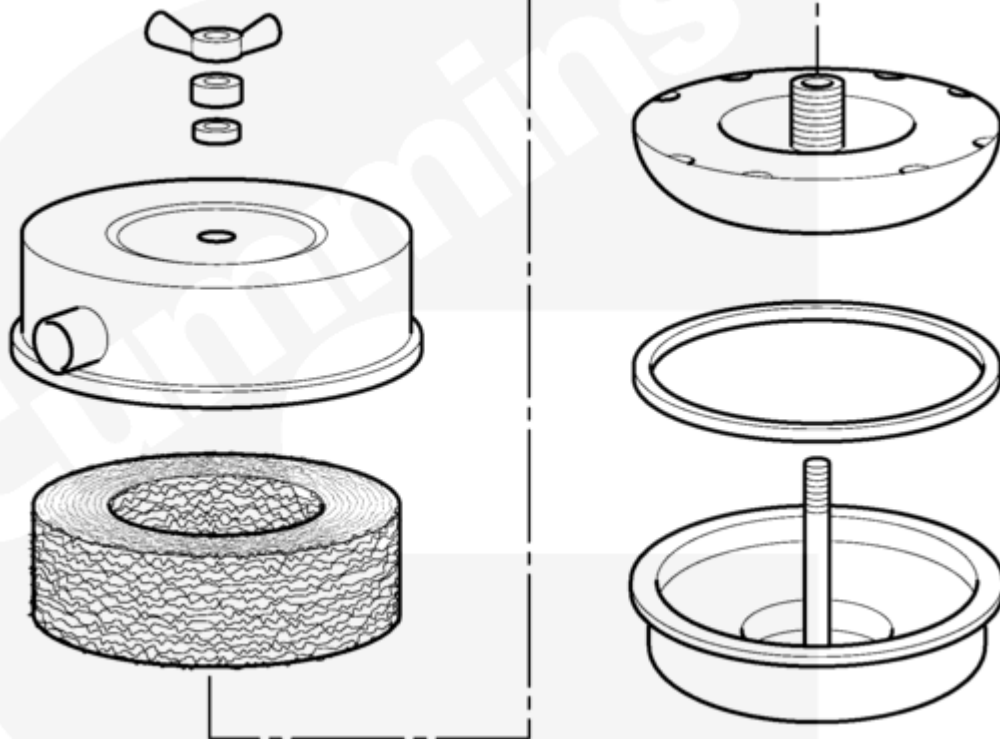
Pressure Checking Air Leaks - Naturally Aspirated Engine

- a. Install the dummy element. The dummy element is a part of the air cleaner checking kit
- b. Make sure that the wing bolt gasket is used with the dummy element, and if required, the inner cover is used
- c. Replace the cover and install the cover retainer assembly from the air cleaner checking kit:
 1. Release the over-center latches
 2. Install the side rod band assembly. Notches fit around the over-center latches so that the band can be tightened around the body
 3. Snap the over-center latches back on the cover
 4. Tighten each of the wing nuts equally until the latches can be "wiggled". Hooks are positioned over the cover lip in the same manner as the latches.
- a. Put 2 psi [14 kPa] maximum air pressure into the cleaner through the ST-439 valve
- b. Use a liquid soap solution to check the following points:
 1. Gasket and plate assembly (if cleaner only is being checked)
 2. Tap the cleaner body
 3. Cover gasket
 4. Upper body reinforcing rim to body
 5. Outlet to body neck
 6. Upper body to lower element retainer seam
 7. Hose and piping if complete system is being checked.
- a. Use Prussian Blue at the following internal joint to see if a good seal is indicated:
 1. Wing bolt gasket washer
 2. Element gasket (seating gasket).



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Oil Bath Crankcase Breather



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Crankcase Breather With Vapor Barrier

Large quantities of dirt can enter an engine through the smallest opening or break.

A crankcase breather performs two functions – it relieves crankcase pressure by venting oil vapors that rise from the crankcase and it filters fresh air entering the engine through the breather.

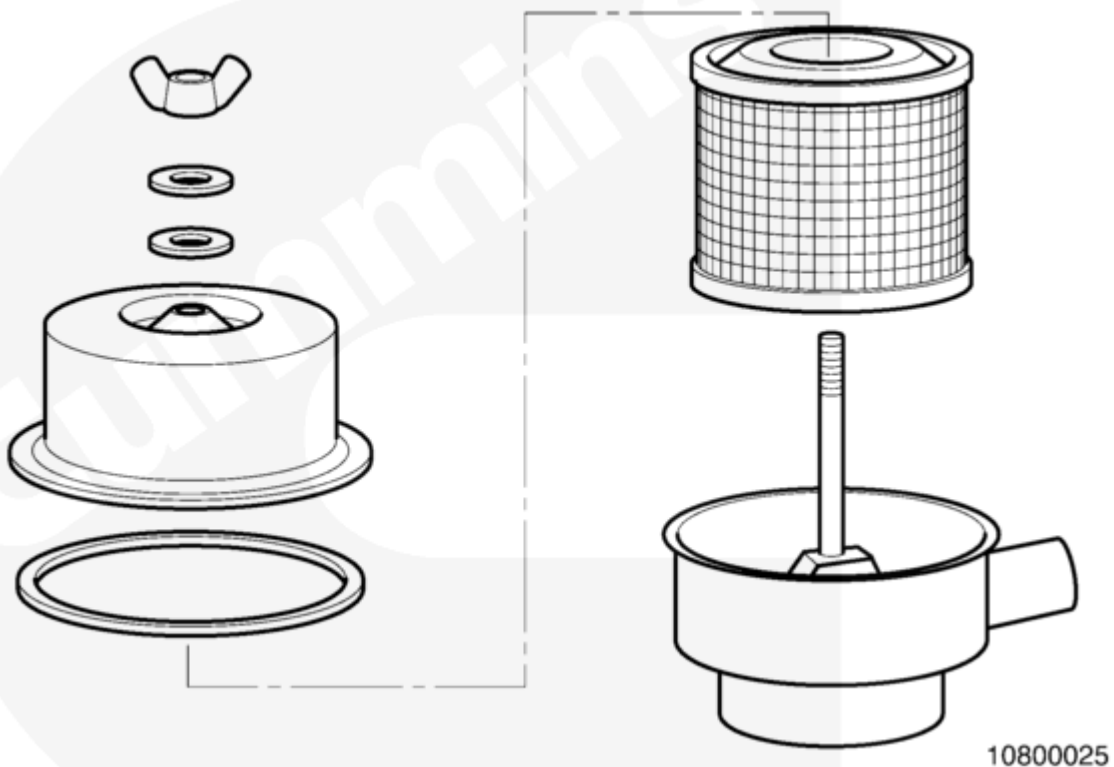
There are two types of crankcase breather loops – close and open. Closed crankcase breathers run the filtered air back to the system where open crankcase breather releases the filtered air to the environment.

Three types of crankcase breather are or have been used on Cummins Engines; oil bath used on older model engines, paper element and mesh element types. The type of breather used depends on whether an engine is turbocharged or naturally aspirated. Currently, new naturally aspirated engines are built without breathers.

An oil bath crankcase breather operates in the same manner as an oil bath type air cleaner. In addition, as oil-laden vapors pass through the breather, they coat the screen with oil and thus increase overall filtering efficiency. The oil bath breather was used and is current for service only on naturally aspirated engines.

Clean the oil bath breather by washing it in cleaning solvent or kerosene and drying with compressed air. Fill the oil cup to the oil level mark with same grade lubricating oil used in the engine.

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Dry Type Crankcase Breather

1. A dry-type paper element crankcase breather was also used on naturally aspirated engines. This breather contains a pleated, porous chemically-treated paper element for crankcase venting and air filtration. A vent tube connected to the breather carries oil vapors away from the engine. Cleaning the elements for reuse is **not** recommended. If breaks appear in the old element, discard it and install a new one.
2. A crankcase breather assembly using mesh as a filtering agent is used on turbocharged engines.

This design features a lower profile, thus allowing additional space for easy servicing and more compact unit for tight installations.

⚠ WARNING ⚠

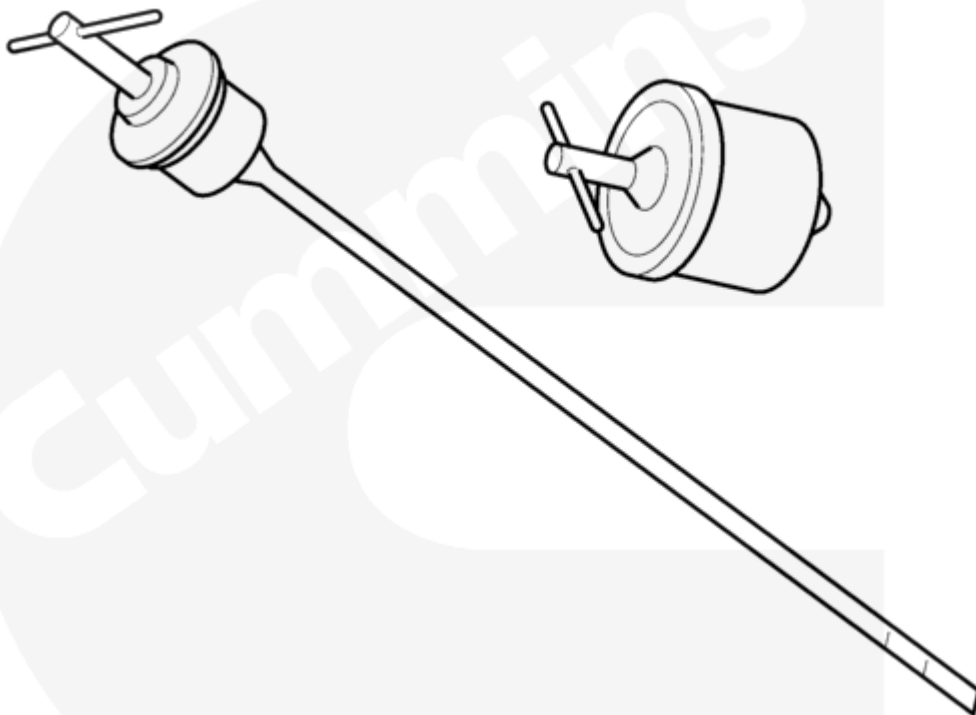
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean this breather by washing the mesh in cleaning solvent and drying it with compressed air.

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Oil Filler Caps And Dipsticks

Dipstick and oil filler caps contain rubber seals which expand on tightening to create an airtight seal that prevents entry of dirt and dust.

Make sure the caps are tightened after each oil check and oil change.

Very little servicing is required to keep dirt from entering through the crankcase breather, oil filler tube, and dipstick tube. However, overlooking these maintenance procedures can mean extra wear on the engine and consequently, shorter engine life.

Clean the air cleaner piping and intake manifold often enough to prevent the buildup of dirt which will restrict the air flow and reduce the air cleaner efficiency.

Keep Intake Air System Efficient By Regular Cleaning

The air cleaner will remove dirt from the air until the cleaner gets loaded; then it restricts air flow and/or lets the dirt go through. The system **must** then be cleaned. See the appropriate Cummins Operation and Maintenance Manuals for the schedule and instructions.

A regular maintenance schedule for the air cleaner system is necessary to make sure that each operation will be performed when it is needed.

Get The Cleanest Air That Is Available

On off-highway applications, it is often necessary to provide an extension to the top of the air cleaner so the air intake will **not** be in the main dust stream. On cab-over trucks the air inlet cap **must** be 14 to 18 inches above the cab roof to minimize water entry.

Make sure the diameter of the extension tube or snorkel is as large as that of the air cleaner inlet flange.

The dust concentration three or four feet above the hood is only a fraction as severe as the concentration at or below hood level.

Common Sense Do's And Don'ts For Engine Intake Air Systems

Don't

- Service the air cleaner when the engine is running
- Judge the plugged condition of an element visually (check by restriction)
- Use oil in dry-type air cleaners
- Operate the engine without an element
- Operate the engine with a damaged filter element, damaged air cleaner or transfer tube
- Let the dust cup or dust receptacle overflow
- Leave new or cleaned elements in open unprotected areas
- Damage the filter element during cleaning
- Operate the engine without the dust cup or a missing actuator valve. (This affects element service life)
- Think that the element is installed properly — know it is
- Over-service the air cleaner
- Drop foreign material into clean air chambers
- Open the air cleaner when dust is blowing around
- Depend upon poor gaskets to give positive sealing
- Open the cover when the engine is running
- Blow through the element by putting it on top of the exhaust pipe and then running the engine.

Do

- Clean off the access cover before removing the elements
- Clean out the clean air chamber if dirt is dropped in
- Inspect the element before reuse

- Check all, and replace the gaskets as required
- Check elements by the restriction method
- Use only proper size approved elements
- Try to keep spare elements on hand
- Check all seals and connections. Remember all intake air **must** go through the air cleaner
- Check for inlet plugging
- Check the dust cup even though a vacuator valve is being used
- Watch for leaks in the exhaust system (carbon entering the element cause short life)
- Keep records of restriction readings and hours or mileage between service intervals
- Check for restriction of all induction parts — elements, cleaner inlet, tube sections or fin areas and inlets
- Check the position of the dust cup to see that it is installed properly and sealing the full 360 degrees to the air cleaner body
- Check that all wing nuts are tight
- Check to see that the elements seal against a good gasket. Some sealing gaskets are on the element itself and some are fixed on the gasket retainer within the air cleaner. It's possible to have an element without a gasket — and an air cleaner body without a gasket — result — no seal
- Check to see that the proper length elements are used. Two elements having the same diameter with different overall lengths will fit in the same air cleaner. The shorter element will stop at the retaining yoke and never make contact with the sealing gasket — result — dirt will bypass the element
- Use your engine Operation and Maintenance Manual for complete servicing instructions.

Document History

Date	Details
XXXX-XX-XX	Module Created
2020-1-17	Adding reference to Service Bulletin 5613318.

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